



The root barks of *Morus alba* and the flavonoid constituents inhibit airway inflammation



Hun Jai Lim^a, Hong-Guang Jin^b, Eun-Rhan Woo^b, Sang Kook Lee^c, Hyun Pyo Kim^{a,*}

^a College of Pharmacy, Kangwon National University, Chuncheon 200-701, South Korea

^b College of Pharmacy, Chosun University, Kwangju 151-759, South Korea

^c College of Pharmacy, Seoul National University, Seoul 151-742, South Korea

ARTICLE INFO

Article history:

Received 11 April 2013

Received in revised form

3 June 2013

Accepted 5 June 2013

Available online 24 June 2013

Keywords:

Morus alba

Prenylated Flavonoid

Lung inflammation

Bronchitis

ABSTRACT

Ethnopharmacological relevance: The root barks of *Morus alba* have been used in traditional medicine as an anti-inflammatory drug, especially for treating lung inflammatory disorders.

Aim of study: To find new alternative agents against airway inflammation and to establish the scientific rationale of the herbal medicine in clinical use, the root barks of *Morus alba* and its flavonoid constituents were examined for the first time for their pharmacological activity against lung inflammation.

Materials and methods: For *in vivo* evaluation, an animal model of lipopolysaccharide-induced airway inflammation in mice was used. An inhibitory action against the production of proinflammatory molecules in lung epithelial cells and lung macrophages was examined.

Results: Against lipopolysaccharide-induced airway inflammation, the ethanol extract of the root barks of *Morus alba* clearly inhibited bronchitis-like symptoms, as determined by TNF- α production, inflammatory cells infiltration and histological observation at 200–400 mg/kg/day by oral administration. In addition, *Morus alba* and their major flavonoid constituents including kuwanone E, kuwanone G and norartocarpone significantly inhibited IL-6 production in lung epithelial cells (A549) and NO production in lung macrophages (MH-S).

Conclusions: Taken together, it is concluded that *Morus alba* and the major prenylated flavonoid constituents have a potential for new agents to control lung inflammation including bronchitis.

© 2013 Elsevier Ireland Ltd. All rights reserved.

1. Introduction

Lung inflammatory disorders include acute and chronic bronchitis. Against these diseases, several classes of drugs such as anti-inflammatory steroids, antitussives, mucolytics and/or bronchodilators are prescribed in clinics to alleviate the various symptoms of airway inflammation. However, it is sometimes difficult to successfully treat these inflammatory disorders, especially chronic bronchitis (one form of chronic obstructive pulmonary diseases, COPD) with currently-available drugs (Jeffery, 2001). Thus, it is necessary to develop new agents in order to fundamentally control these diseases and to have alternative drug (s) showing different action mechanism(s). Among the many efforts to find new drug candidates, a search for active constituents from various plant products has been continued.

The root barks of *Morus alba* L. have been widely used in traditional medicine for treating various inflammatory conditions, including acute and chronic bronchitis in North-East

Asia (Bae, 2000). Previous phytochemical studies on this plant reported the isolation of various flavonoids. Moreover, it is well-known that the main constituents of this plant material are the isoprene substituted flavonoids, prenylated flavonoids (Nomura, 2001). Previously, the methanol and ethylacetate extracts of the root barks of *Morus alba* were reported to show anti-inflammatory activity (Hong et al., 2002; Chao et al., 2009). The prenylated flavonoids such as morusin, kuwanon C and kuwanon G isolated from *Morus alba*, have also been shown to possess various anti-inflammatory activities (Cheon et al., 2000; Yang et al., 2011). However, despite the frequent use on lung inflammation-related disorders, the pharmacological evaluation of the total extract of *Morus alba* as well as its major constituents on airway inflammation has not been carried out to date. Therefore, in the present study, the inhibitory action of the ethanol extract of *Morus alba* was examined using a lipopolysaccharide (LPS)-induced airway inflammation model in mice to provide the scientific rationale for the clinical use. The main flavonoid constituents were isolated from the root barks of *Morus alba* and their effects on inflammatory responses of lung epithelial cells and alveolar macrophages were investigated to prove the potential for new therapeutic agents.

* Corresponding author. Tel.: +82 33 2506915; fax: +82 33 2557865.
E-mail address: hpkim@kangwon.ac.kr (H.P. Kim).

2. Materials and methods

2.1. Chemicals

2-Amino-5,6-dihydro-6-methyl-4 H-1,3-thiazine hydrochloride (AMT) was purchased from Tocris Cookson Ltd. (UK). 3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), dexamethasone, IL-1 β and LPS (*Escherichia coli* 0127:B8) were purchased from Sigma Chem. (St. Louis, MO). MEM and other cell culture reagents including FBS were products of Gibco BRL (Grand Island, NY). Protein assay kit was purchased from Bio-Rad Lab. (Hercules, CA).

2.2. Animals

Male ICR mice (male, 18–22 g, specific pathogen-free) were obtained from Central Exp. Animal Ltd. (Korea). Animals were fed with standard lab. chow and water *ad libitum*. The animals were maintained in animal facility (KNU) at 20–22 °C under 40–60% relative humidity and a 12 h/12 h (light/dark) cycle for at least 7 days prior to the experiment. The experimental design using the animals was approved by the local committee for animal experimentation, KNU (KIACUC-12-0012). In addition, the ethical guideline described in the Korean Food and Drug Administration guide for the care and use of laboratory animals was followed throughout the experiments.

2.3. Plant materials

The root barks of *Morus alba* L. (Moraceae) were collected in Geongju, Geongbuk province, Korea, and authenticated by Prof. J. H. Lee (Dongguk University, Gyeongju, Korea). The voucher specimen (CSU-1048-17) were deposited in the Herbarium of the College of Pharmacy, Chosun University.

2.4. Extraction and isolation of kuwanon E, kuwanon G, and norartocarpanone

The dried root barks of *Morus alba* (1 kg) were extracted with 70% ethanol under reflux. Evaporation under reduced pressure gave the dried extract (MA, 170.3 g), which was used throughout the pharmacological study. In addition, for isolating major flavonoid constituents, the dried root barks of *Morus alba* (12 kg) were extracted three times with methanol under reflux and 1511.6 g of residue was produced. The methanol extract was suspended in water and was then partitioned sequentially with equal volumes of dichloromethane (CH₂Cl₂), ethyl acetate (EtOAc), and *n*-butanol (*n*-BuOH). Each fraction was evaporated under reduced pressure to yield the residues of CH₂Cl₂ (318.2 g), EtOAc (192.2 g), *n*-BuOH (182.4 g), and water (534.3 g) extract. The EtOAc fraction (53 g) was chromatographed over a silica gel column using a gradient solvent system of *n*-hexane-EtOAc (2:1 $\rightarrow$ 1:8, EtOAc, MeOH) to give 5 subfractions (E1–E5). Subfraction E2 was subjected to MCI gel column chromatography (CC) eluting with a gradient solvent system of MeOH:H₂O (1:1 $\rightarrow$ 3:1) to yield 16 subfractions (E21–E216). Subfraction E22 was purified by LiChroprep RP 18 CC (MeOH:H₂O, 2:3 $\rightarrow$ 1:1) to give norartocarpanone (3.42 mg). Subfraction E24 was subjected to silica gel CC (CHCl₃:MeOH:H₂O, 12:1:0.1 $\rightarrow$ 8:1:0.1) to give 6 fractions (E241–E246). Subfraction E245 was purified by LiChroprep RP 18 CC (MeOH:H₂O, 1:1) to yield kuwanon G (1011 mg). Subfraction E27 was purified repeated silica gel CC (CHCl₃:MeOH:H₂O, 25:1:0.1 $\rightarrow$ 10:1:0.1) and LiChroprep RP 18 CC (MeOH:H₂O, 2:1) to yield kuwanon E (43.7 mg) (Fig. 1). The physico-chemical data including ¹H NMR, ¹³C NMR, and HSQC of these compounds were identical with those reported

in the literature (Hano et al., 1984; Nomura et al., 1980; Nomura and Fukai, 1981).

Kuwanon E: ¹H-NMR (Acetone-*d*₆, 600 MHz): δ 7.20 (1H, s, H-6'), 6.50 (1H, s, H-3'), 5.95 (1H, d, *J*=1.8 Hz, H-6), 5.94 (1H, d, *J*=1.8 Hz, H-8), 5.70 (1H, dd, *J*=3.0, 13.8 Hz, H-2), 5.35 (1H, dt, *J*=1.2, 7.2 Hz, H-2''), 5.11 (1H, tt, *J*=1.2, 7.2 Hz, H-7''), 3.27 (2H, d, *J*=7.2 Hz, H-1'), 3.20 (1H, dd, *J*=13.8, 17.4 Hz, Ha-3), 2.69 (1H, dd, *J*=3.0, 17.4 Hz, Hb-3), 2.09 (2H, m, H-6''), 2.02 (2H, t, *J*=7.2 Hz, H-5''), 1.71 (3H, s, H-4''), 1.62 (3H, s, H-9''), 1.57 (3H, s, H-10''); ¹³C-NMR (Acetone-*d*₆, 150 MHz): δ 197.8 (C-4), 167.3 (C-7), 165.4 (C-8a), 164.9 (C-5), 156.7 (C-4'), 154.2 (C-2'), 135.9 (C-3'), 131.7 (C-8''), 129.0 (C-6'), 125.2 (C-7''), 124.1 (C-2''), 120.2 (C-5'), 116.9 (C-1'), 103.4 (C-3'), 103.2 (C-4a), 96.7 (C-8), 95.9 (C-6), 75.5 (C-2), 42.7 (C-3), 40.5 (C-5''), 28.4 (C-1''), 27.5 (C-6''), 25.9 (C-9''), 17.8 (C-10''), 16.3 (C-4'').

Kuwanon G: ¹H-NMR (Acetone-*d*₆, 300 MHz): δ 7.41 (1H, d, *J*=8.0 Hz, H-6' or H-27), 7.29 (1H, d, *J*=8.0 Hz, H-6' or H-27), 6.78 (1H, d, *J*=8.0 Hz, H-33), 6.67 (1H, d, *J*=2.0 Hz, H-3'), 6.55 (1H, dd, *J*=2.0, 8.0 Hz, H-5'), 6.21 (1H, d, *J*=2.0 Hz, H-30), 6.08 (1H, dd, *J*=2.0, 8.0 Hz, H-32), 6.03 (1H, d, *J*=2.0 Hz, H-24), 5.98 (1H, s, H-6), 5.93 (1H, dd, *J*=2.0, 8.0 Hz, H-26), 4.95–5.40 (2H, m, H-10 and H-15), 4.30–4.70 (2H, m, H-14 and H-20), 3.30–3.90 (1H, m, H-19), 3.17 (2H, d, *J*=7.0 Hz, H-9), 1.80–2.20 (1H, m, H-18), 1.62 (3H, s, H-12), 1.52 (3H, s, H-16), 1.48 (3H, s, H-13); ¹³C-NMR (DMSO-*d*₆, 75.5 MHz): δ 208.1 (C-21), 181.7 (C-4), 164.2 (C-23), 164.2 (C-25), 161.3 (C-7), 160.83 (C-4'), 160.3 (C-8a), 159.2 (C-2), 156.3 (C-2'), 155.8 (C-29), 155.8 (C-31), 155.2 (C-5), 132.8 (C-16), 132.4 (C-33), 131.2 (C-11), 131.2 (C-6'), 130.8 (C-27), 123.2 (C-15), 121.8 (C-10), 120.7 (C-28), 119.7 (C-3), 114.0 (C-22), 111.4 (C-1'), 107.2 (C-26), 106.7 (C-8), 106.7 (C-32), 106.7 (C-5'), 103.7 (C-4a), 102.6 (C-24), 102.6 (C-3'), 101.9 (C-30), 97.5 (C-6), 45.8 (C-20), 38.3 (C-18), 38.3 (C-19), 25.4 (C-12), 23.5 (C-9), 22.9 (C-14), 22.5 (C-17), 17.3 (C-13).

Norartocarpanone: ¹H-NMR (Acetone-*d*₆, 600 MHz) δ : 7.31 (1H, d, *J*=8.4 Hz, H-6'), 6.47 (1H, d, *J*=1.8 Hz, H-3'), 6.43 (1H, dd, *J*=1.8, 8.4 Hz, H-5'), 5.96 (1H, d, *J*=2.4 Hz, H-8), 5.94 (1H, d, *J*=2.4 Hz, H-6), 5.70 (1H, dd, *J*=3.0, 13.2 Hz, H-2), 3.17 (1H, dd, *J*=13.2, 17.4 Hz, H-3), 2.71 (1H, dd, *J*=3.0, 17.4 Hz, H-3); ¹³C-NMR (Acetone-*d*₆, 150 MHz) δ : 197.7 (C-4), 167.7 (C-7), 165.4 (C-8a), 164.9 (C-5), 159.6 (C-2'), 156.4 (C-4'), 129.1 (C-6'), 117.5 (C-1'), 107.8 (C-5'), 103.5 (C-3'), 103.1 (C-4a), 96.8 (C-6), 95.9 (C-8), 75.4 (C-2), 42.7 (C-3).

2.5. LPS-induced airway inflammation in mice

Mice were divided into five groups, including the control, LPS-treated, LPS/MA (200, 400 mg/kg/day)-treated and LPS/dexamethasone-treated (*n*=14). The test compounds, including the reference drug, were dissolved in 0.3% carboxymethylcellulose (CMC) and were orally administered for three consecutive days. The control and LPS treatment groups also received the CMC solution. For inducing bronchitis, LPS (*Escherichia coli* 0127:B8, 800 μ g/ml, saline) was administered intranasally to mice (10 μ l/mouse, 5 times) 1 h after the final oral treatment of the test compounds. At 4 h after LPS treatment, 5 mice per group were sacrificed and bronchoalveolar lavage fluid (BALF) was collected via intratracheal cannulation after 700 μ l of saline was administered 3 times. BALFs collected were approximately 2000 μ l/mouse. From BALF, TNF- α concentration was measured with an ELISA kit (eBioscience) according to the manufacturer's recommendation. At 16 h after LPS treatment, mice were sacrificed (*n*=7), and BALF was obtained. From BALF, the total cell number was counted with a haemocytometer, and the cells were differentially counted with FACS (BD Bioscience). For histology and biochemical analysis, the remaining mice (*n*=2) were sacrificed and lung tissues were excised. Histology was carried out using conventional methods of H&E staining.

2.6. A549 cell culture and measurement of IL-6 concentration

A549 cells, a human lung epithelial cell line, obtained from American type culture collection (ATCC, Rockville, VA) were cultured with RPMI supplemented with 10% FBS, 1% L-glutamine and 1% antibiotics (100 U/ml penicillin and 100 µg/ml streptomycin) in a 5% CO₂ atmosphere at 37 °C. After pre-incubation for 24 h, IL-1β (10 ng/ml) was added simultaneously with/without test compounds. Four hours later, media was collected and the concentration of IL-6 was determined from the media with ELISA kit (eBioscience) according to the manufacturer's recommendation. The cell viability was checked using an MTT bioassay as previously described (Mossmann, 1983).

The test compounds including reference drug were dissolved in DMSO and properly diluted with complete RPMI media. The final concentration of DMSO in the cell culture including MH-S cells below was adjusted to 0.1% (v/v), and this concentration of DMSO did not affect the cell viability and the levels of IL-6 production (data not shown).

2.7. MH-S cell culture and measurement of nitric oxide (NO) concentration

MH-S cells, a mouse alveolar macrophage cell line, obtained from ATCC were cultured with RPMI supplemented with 10% FBS, 1% L-glutamine and 1% antibiotics (100 U/ml penicillin and 100 µg/ml streptomycin) in a 5% CO₂ atmosphere at 37 °C. As previously described (Ko et al., 2011), cells were plated in 96-well plates (2×10^5 cells/well). After pre-incubation for 24 h, test compounds and LPS (0.1 µg/ml) were added and the cells were incubated for 24 h unless otherwise specified. Media was collected and NO

concentration was determined. To assess NO production, the stable conversion product of NO, nitrite (NO₂⁻), was measured using Griess reagent and the optical density was determined at 550 nm.

2.8. Statistical analysis

Experimental values were represented as arithmetic mean ± SD. One way ANOVA followed by Dunnett's test was used to determine the statistical significance.

3. Results

3.1. Effects of MA on LPS-induced airway inflammation in mice

A mouse model of airway inflammation induced by intranasal treatment of LPS is frequently used for establishing the effects of various compounds on airway inflammation including symptoms of bronchitis and COPD (Chapman et al., 2007). LPS treatment to mice provokes inflammatory responses of inflammatory cell recruitment, elevated levels of proinflammatory cytokines and histologic changes in the lung tissue. In our experiment, the intranasal administration of LPS produced similar changes in BALF and in histology, as expected. TNF-α concentration greatly increased to 1.33 ± 0.54 ng/ml from the basal level of 0.37 ± 0.01 ng/ml ($n=5$) (Fig. 2A). The total cells in BALF significantly increased (approximately 3.6-fold). The cells gathered by LPS treatment in BALF include infiltrated neutrophils, damaged and detached residents, and/or infiltrated macrophages, and dendritic cells, as revealed by FACS analysis (Fig. 2B). Histological changes of hyperplasia of the epithelial cell layer and narrowed alveolar space were also noticed (Fig. 2C). All these responses indicated that the intranasal administration of LPS produced bronchitis-like responses in mice. Under this condition, the oral administration of MA clearly reduced these parameters. Particularly, MA (400 mg/kg/day) significantly and strongly reduced the numbers of total cells in BALF (51.2% inhibition), numbers of alveolar as well as interstitial macrophages (70.9% and 43.8% inhibition, respectively) and dendritic cells (52.5% inhibition). It also reduced the infiltration of neutrophils (69.1% inhibition), but not by a statistically significant degree. MA (400 mg/kg/day) did not exert a significant reduction of TNF-α concentration (28.7% reduction) in BALF, while MA (200 mg/kg/day) strongly reduced the production of TNF-α (72.4% reduction). The results of histology also support the notion that MA possesses anti-bronchitis activity. MA (200 and 400 mg/kg/day) did inhibit epithelial hyperplasia and narrowing alveolar space. As expected, dexamethasone, which was used as a reference drug, showed strong inhibitory action on all these biomarkers, except neutrophil infiltration. Dexamethasone slightly reduced neutrophil infiltration (12.4%) without statistical significance.

3.2. Effects of MA and the flavonoid constituents on IL-6 production in lung epithelial cells

Since MA possesses anti-bronchitis activity as above, the major flavonoid constituents were isolated and the constituents were examined for anti-inflammatory activity in lung cells. The constituents isolated were kuwanon E, kuwanon G and norartocarpanone.

IL-1β treatment to A549 cells provokes IL-6 production of 0.73 ± 0.03 ng/ml ($n=3$) for 4 h (Fig. 3). The basal level was 0.02 ± 0.02 ng/ml. The extract (MA) clearly inhibited IL-6 production at 300 µg/ml (54.8% inhibition). When the constituents isolated (10–100 µM) were incubated with IL-1β, they showed an inhibition of IL-6 production at the concentration ranges of 50–100 µM depending on the compounds. In particular, kuwanone E showed a concentration-dependent inhibition at 20–100 µM

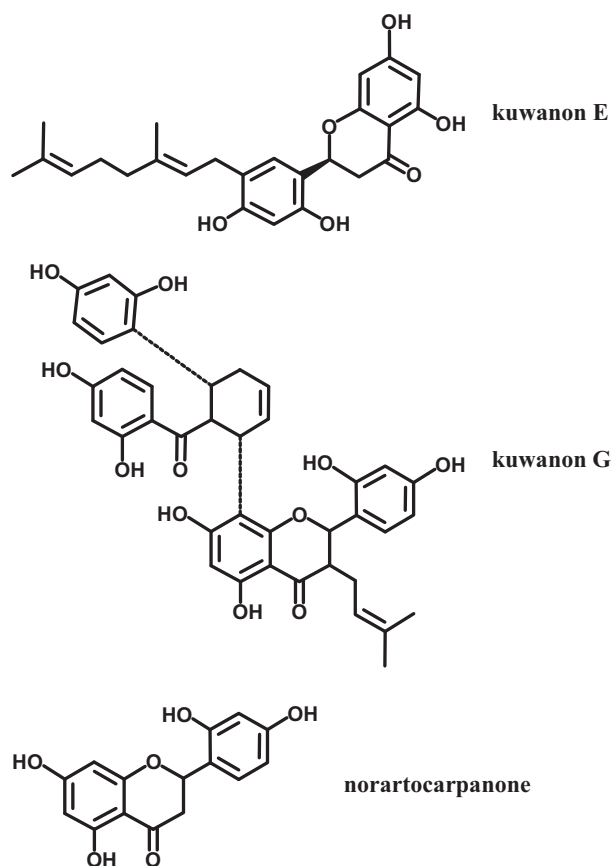


Fig. 1. The chemical structures of the flavonoids isolated in this study.

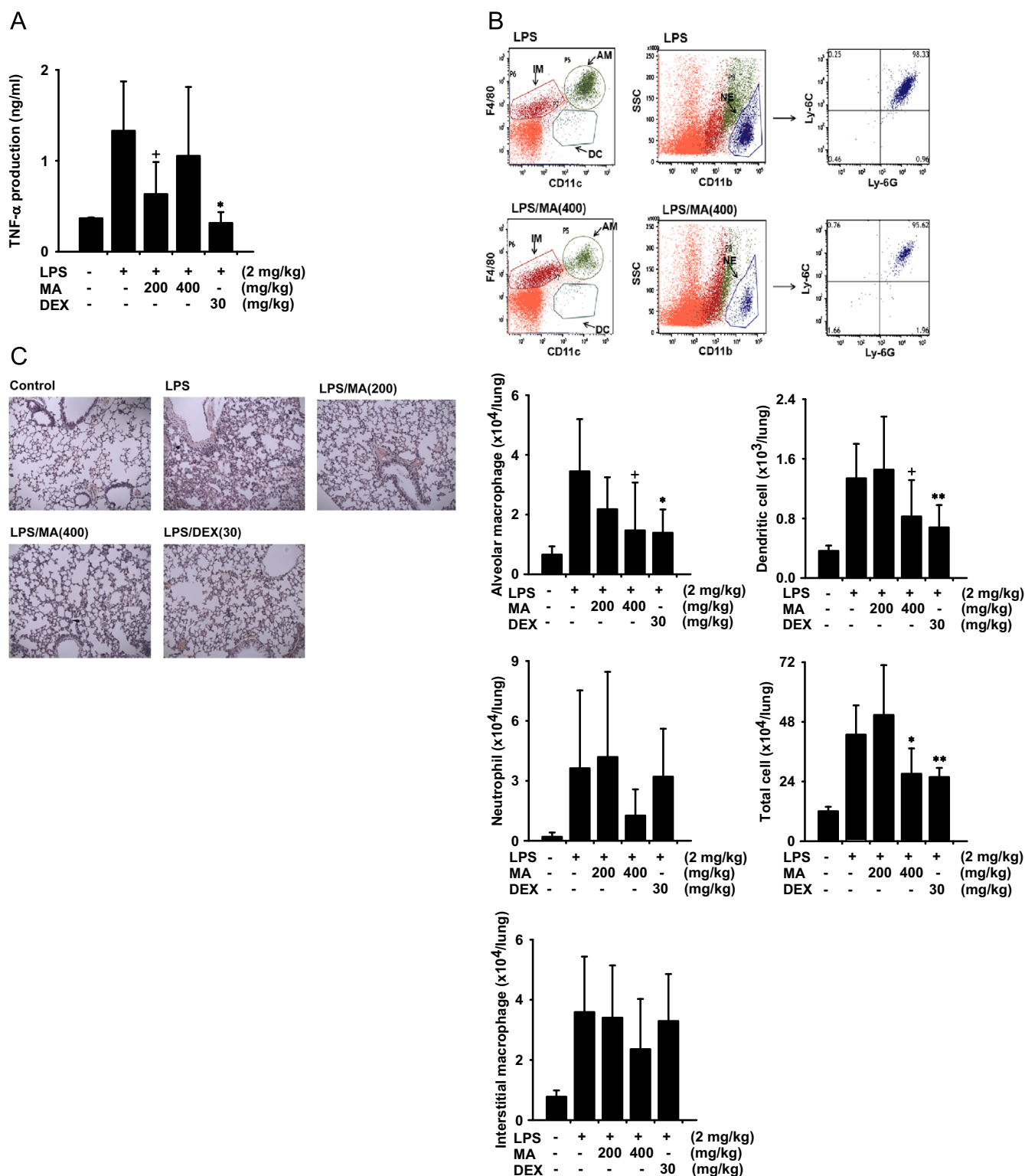


Fig. 2. Inhibition of *Morus alba* (MA) on LPS-induced airway inflammation in mice. (A) Effects on TNF- α production in BALF, LPS was administered to mice intranasally. Four hours later, mice were sacrificed and TNF- α concentration was measured. DEX: dexamethasone, +: $P < 0.1$, *: $P < 0.05$, and **: $P < 0.01$, Significantly different from the LPS-treated group ($n=5$). (B) Effects on the cell numbers in BALF, Total cell numbers were counted using the haemocytometer. FACS was used to differentiate between each cell type of inflammation-related cells. DEX: dexamethasone, +: $P < 0.1$, *: $P < 0.05$, and **: $P < 0.01$, Significantly different from the LPS-treated group ($n=7$). (C) Histological observation of lung tissues (H&E staining), Here represented one sample of two sets stained, $100\times$.

(IC₅₀=47.5 μ M). None of the compounds exerted a cytotoxic effect on A549 cells, as determined by MTT assay under experimental conditions (data not shown). Dexamethasone (10 μ M), which was used as a reference drug showed potent inhibition (87.5%), as expected.

3.3. Effects of MA and the flavonoid constituents on NO production in alveolar macrophages

Likewise, LPS treatment to MH-S cells induced inducible NO synthase (iNOS) producing high amounts of NO (Ko et al.,

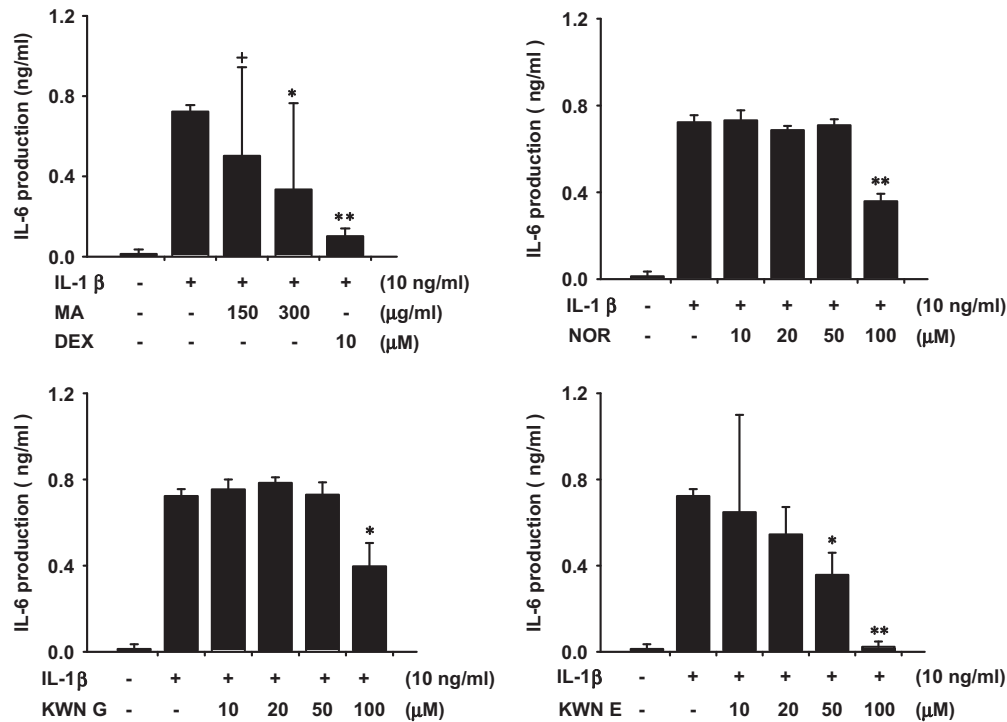


Fig. 3. Inhibition of *Morus alba* and the flavonoid constituents on IL-6 production in A549 cells. NOR: norartocarpanone, KWN E: kuwanon E, KWN G: kuwanon G, DEX: dexamethasone, CT: cytotoxic, +: $P < 0.1$, *: $P < 0.05$, and **: $P < 0.01$. Significantly different from the LPS-treated group ($n=3$).

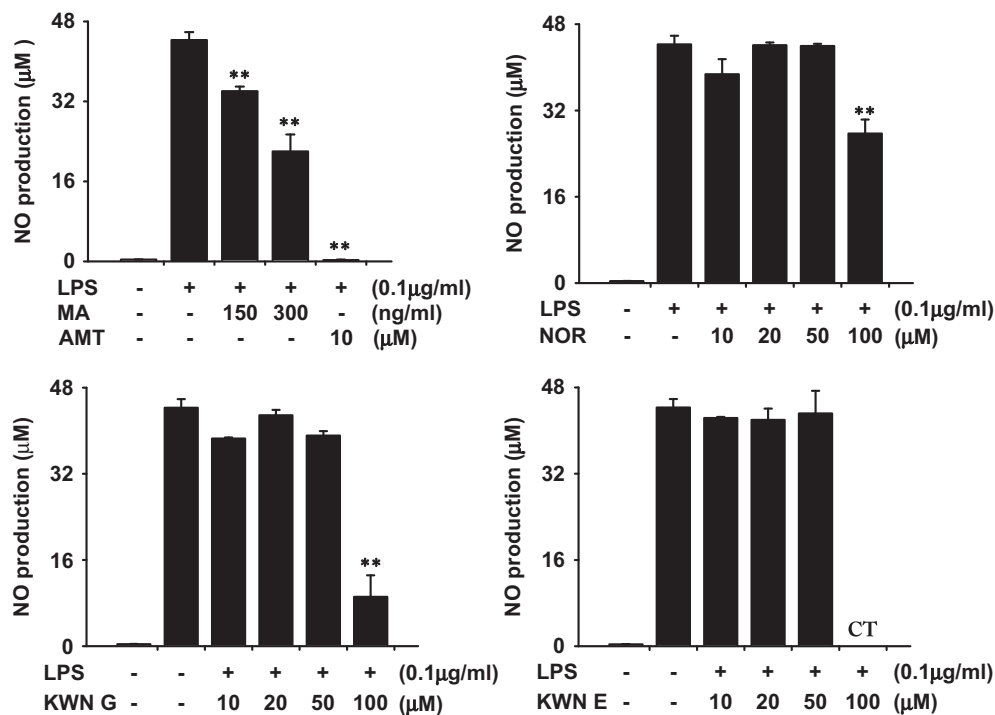


Fig. 4. Inhibition of *Morus alba* and the flavonoid constituents on iNOS-catalyzed NO production in MH-S cells. NOR: norartocarpanone, KWN E: kuwanon E, KWN G: kuwanon G, DEX: dexamethasone, CT: cytotoxic, +: $P < 0.1$, *: $P < 0.05$, and **: $P < 0.01$. Significantly different from the LPS-treated group ($n=3$).

2011). In this experiment, NO concentration in the media increased to 44.3 ± 1.6 μM by LPS treatment for 24 h, while the basal level of NO was 0.4 ± 0.1 μM (Fig. 4). Under this condition, the extract and the constituents inhibited NO production. MA showed 23.3% and 50.7% inhibition at 150 and

300 μg/ml, respectively. The flavonoids, kuwanon G and norartocarpanone, significantly inhibited NO production at 100 μM (79.9% and 37.7%, respectively). AMT (10 μM), which was used as a reference agent showed potent inhibition (99.9%) as expected. All compounds did not exert a cytotoxic effect on

MH-S cells as judged by MTT assay under experimental conditions, except kuwanone E (24.8% viability reduction at 100 μ M).

4. Discussion

The present investigation has clearly shown that the root barks of *Morus alba* possess the inhibitory action on airway inflammation *in vitro* and *in vivo*. The flavonoid constituents showed a similar effect in lung cells, suggesting that these constituents may contribute, at least in part, to the inhibitory action of *Morus alba*.

In vivo study has shown that MA reduced total cell numbers and TNF- α concentration in BALF provoked by LPS treatment. TNF- α is one of the most prominent inflammatory cytokines to produce inflammatory response in lung (Mueller et al., 1996). By LPS treatment, TNF- α concentration in the lung increased rapidly at 4 h, presumably via a production by lung epithelial cells and macrophages. However, at 16 h after LPS treatment, BALF no longer contained measurable amounts of TNF- α (data not shown), suggesting that TNF- α production in the lung rapidly decreased to the baseline level in response to LPS stimulation. On the other hand, the inflammation-related cells, including neutrophils, dendritic cells and macrophages, were infiltrated to the lung, gradually increased and reached a peak at about 16–24 h after LPS treatment (data not shown). Thus, in our study, TNF- α level was checked at 4 h, and the total cells in BALF were measured at 16 h after LPS treatment. Under these conditions, MA showed strong inhibition of TNF- α production at 200 mg/kg/day, as well as cell recruitment in the lung at 400 mg/kg/day. However, at present, it is not understood why the high-dose treatment of MA (400 mg/kg/day) inhibited TNF- α production in BALF to a lesser degree than the low-dose treatment of MA (200 mg/kg/day).

In vitro study also revealed that MA inhibited IL-6 production in lung epithelial cells and NO production in lung macrophages. Since these molecules (IL-6 and NO) are important biomarkers in inflammatory disorders including lung inflammation, it is thought that MA possesses broad spectrum inhibitory action mechanisms on lung inflammation. The major flavonoid constituents also possessed a similar inhibitory activity on these biomarkers.

It is well-known that flavonoids are major anti-inflammatory principles in *Morus alba*, particularly prenylated flavonoids (Nomura, 2001). Kuwanone G and norartocarpone were found to inhibit iNOS-catalyzed NO production in the present investigation. The NO inhibitory action of kuwanon G is well correlated with the previous findings (Cheon et al., 2000) that some prenylated flavonoid possess NO inhibitory action on LPS-treated RAW 264.7 cells, a mouse macrophage cell line. In addition, kuwanone E and norartocarpone were found to inhibit IL-6 production in lung epithelial cells. These findings may confirm again an important role of the flavonoids in MA for exerting its anti-inflammatory action, especially, on lung inflammation. The similar findings were also obtained by our previous studies demonstrating that the major active constituents of *Betula papyrifera* (Moraceae), anti-bronchitis herbal medicine, are the prenylated flavonoids such as papyriflavonol A and broussonchalcone A (Chi et al., 2001; Cheng et al., 2001). *Morus mongolica*, one of the Moraceae family, also contains prenylated flavonoids as active principles, showing anti-inflammatory activity (Chi et al., 2001). Thus, it is strongly suggested that flavonoid constituents may contribute to the anti-bronchitis activity of *Morus alba*.

Against respiratory tract-related inflammatory disorders including bronchitis, several plant-based anti-inflammatory drugs are currently prescribed in European and Asian countries. Examples include the extracts of *Hedera helix* (Guo et al., 2006), *Echinacea purpurea* (Sharma et al., 2006; Agbabiaka et al., 2008)

and *Pelargonium sidoides* (Matthys and Funk, 2008). Recently, one of the medicinal plants in the same Moraceae family, *Brossonetia papyrifera*, was found to possess anti-bronchitis activity (Ko et al., 2011). The present study was carried out to find plant-based products having a potential for new and alternative medicine. The extract from the root barks of *Morus alba* and several flavonoid constituents were revealed to have such an activity. Thus it is suggested that MA may be a new candidate for treating lung inflammation-related disorders. Previously, *Morus alba* was reported to exert antiasthmatic effects (Kim et al., 2011) and to have anti-allergic properties (Chai et al., 2005). And one herbal combination, Xia-Bai-San having the water extracts of *Morus alba*, *Lycium chinensis*, *Glycyrrhiza uralensis*, *Oryza sativa*, was previously reported to inhibit *in vivo* lung inflammation (Yeh et al., 2006; Lee et al., 2009). Additionally, the extract of *Morus alba* was reported to inhibit IL-8 and ICAM-1 production from A549 cells (Lee et al., 2009). However, despite all these previous findings, the effect of *Morus alba* alone on *in vivo* lung inflammation has not been demonstrated. To the best of our knowledge, the present study is the first demonstration of *in vivo* inhibitory action of *Morus alba* on lung inflammation.

5. Conclusion

All results of the present study demonstrated that the root barks of *Morus alba* inhibited airway inflammation *in vivo* and *in vitro*. And the several flavonoid constituents such as kuwanon E, kuwanon G and norartocarpone were found to inhibit the production of inflammatory biomarkers in lung cells, lung epithelial cells (A549) and alveolar macrophage cells (MH-S), and they may contribute to the inhibitory action of *Morus alba*. Taken together, it is suggested that *Morus alba* and the major flavonoid constituents have potential as new therapeutic agents for airway inflammation.

Acknowledgements

This study was financially supported by a grant (12172KFDA989) from Korea Food and Drug Administration in 2012, and post-BK21 project from the Ministry of Science and Education. Bioassay facility of New Drug Development Institute (KNU) was used and greatly acknowledged.

References

- Agbabiaka, T.B., Guo, R., Ernst, E., 2008. *Pelargonium sidoides* for acute bronchitis: a systematic review and meta-analysis. *Phytomedicine* 15, 378–385.
- Bae, K., 2000. The medicinal plants of Korea. Kyo-Hak Publishing Co., Seoul, Korea p. 73.
- Chai, O.H., Lee, M.S., Han, E.H., Kim, H.T., Song, C.H., 2005. Inhibitory effects of *Morus alba* on compound 48/80-induced anaphylactic reactions and anti-chicken gamma globulin IgE-mediated mast cell activation. *Biological and Pharmaceutical Bulletin* 28, 1852–1858.
- Chao, W.W., Kuo, Y.H., Li, W.C., Lin, B.F., 2009. The production of nitric oxide and prostaglandin E_2 in peritoneal macrophages is inhibited by *Andrographis paniculata*, *Angelica sinensis* and *Morus alba* ethyl acetate fractions. *Journal of Ethnopharmacology* 122, 68–75.
- Chapman, R.W., Minnicozzi, M., Celly, C.S., Phillips, J.E., Kung, T.T., Hipkim, R.W., Fan, X., Rindgen, D., Deno, G., Bond, R., Gonsiorek, W., Billah, M.M., Fine, J.S., Hey, J. A., 2007. A novel, orally active CXCR1/2 receptor antagonist, Sch527123, inhibits neutrophil recruitment, mucus production, and goblet cell hyperplasia in animal models of pulmonary inflammation. *Journal of Pharmacological and Experimental Therapeutics* 322, 486–493.
- Cheng, Z.J., Lin, C.N., Hwang, T.L., Teng, C.M., 2001. Broussonchalcone A, a potent antioxidant and effective suppressor of inducible nitric oxide synthase in lipopolysaccharide-activated macrophages. *Biochemical Pharmacology* 61, 939–946.
- Cheon, B.S., Kim, Y.H., Son, K.H., Chang, H.W., Kang, S.S., Kim, H.P., 2000. Effects of prenylated flavonoids and biflavonoids on lipopolysaccharide-induced nitric oxide production from the mouse macrophage cell line RAW 264.7. *Planta Medica* 66, 596–600.

- Chi, Y.S., Jong, H.G., Son, K.H., Chang, H.W., Kang, S.S., Kim, H.P., 2001. Effects of naturally occurring prenylated flavonoids on arachidonic acid metabolizing enzymes: Cyclooxygenases and lipoxygenases. *Biochemical Pharmacology* 62, 1185–1191.
- Guo, R., Pittler, M.H., Ernst, E., 2006. Herbal medicines for the treatment of COPD: a systematic review. *European Respiratory Journal* 28, 330–338.
- Hano, Y., Hirakura, K., Nomura, T., Terada, S., Fukushima, K., 1984. Structures of two new natural Diels-Alder adducts, kuwanons N and O. *Planta Medica* 50, 127–130.
- Hong, C.H., Hur, S.K., Oh, O.J., Kim, S.S., Nam, K.A., Lee, S.K., 2002. Evaluation of natural products on inhibition of inducible cyclooxygenase (COX-2) and nitric oxide synthase (iNOS) in cultured mouse macrophage cells. *Journal of Ethnopharmacology* 83, 153–159.
- Jeffery, P.K., 2001. Remodeling in asthma and chronic obstructive lung disease. *American Journal of Respiratory and Critical Care Medicine* 164, S28–S38.
- Kim, H.J., Lee, H.J., Jeong, S.J., Lee, H.J., Kim, S.H., Park, E.J., 2011. Cortex *Mori Radicis* extract exerts antiasthmatic effects via enhancement of CD(4)CD25(+)Foxp3(+) regulatory T cells and inhibition of Th2 cytokines in a mouse asthma model. *Journal of Ethnopharmacology* 138, 40–46.
- Ko, H.J., Jin, J.H., Kwon, O.S., Kim, J.T., Son, K.H., Kim, H.P., 2011. Inhibition of experimental lung inflammation and bronchitis by phytoformula containing *Broussonetia papyrifera* and *Lonicera japonica*. *Biomolecules and Therapeutics* 19, 324–330.
- Lee, K.-H., Yeh, M.-H., Kao, S.-T., Hung, C.-M., Chen, B.-C., Liu, C.-J., Yeh, C.-C., 2009. Xia-Bai-San inhibits lipopolysaccharide-induced activation of intercellular adhesion molecule-1 and nuclear factor-kappa B in human lung cells. *Journal of Ethnopharmacology* 124, 530–538.
- Matthys, H., Funk, P., 2008. EPs 7630 improves acute bronchitic symptoms and shortens time to remission. Results of a randomized, double-blind, placebo-controlled, multicenter trial. *Planta Medica* 74, 686–692.
- Mossman, T., 1983. Rapid colorimetric assay for cellular growth and survival: application to proliferation and cytotoxic assays. *Journal of Immunological Methods* 65, 55–63.
- Mueller, R., Chanez, P., Campbell, A.M., Bousquet, J., Heusser, C., Bullock, G.R., 1996. Different cytokine patterns in bronchial biopsies in asthma and chronic bronchitis. *Respiratory Medicine* 90, 79–85.
- Nomura, T., 2001. Chemistry and biosynthesis of prenylflavonoids. *Yakugaku Zasshi* 121, 535–556.
- Nomura, T., Fukai, T., 1981. Constituents of the cultivated mulberry tree VII. Isolation of three new isoprenoid-flavanones, kuwanons D, E, and F from the root bark of *Morus alba* L. *Planta Medica* 42, 79–88.
- Nomura, T., Fukai, T., Narita, T., 1980. Hypotensive constituent, kuwanon H, a new flavone derivative from the root bark of the cultivated mulberry tree (*Morus alba* L.). *Heterocycles* 14, 1943–1951.
- Sharma, M., Arnason, J.T., Burt, A., Hudson, J.H., 2006. Echinacea extracts modulate the pattern of Chemokine and cytokine secretion in rhinovirus-infected and uninfected epithelial cells. *Phytotherapy Research* 20, 147–152.
- Yang, Z.G., Matsuzaki, K., Takamatsu, S., Kitanaka, S., 2011. Inhibitory effects of constituents from *Morus alba* var. *multicaulis* on differentiation of 3T3-L1 cells and nitric oxide production in RAW 264.7 cells. *Molecules* 16, 6010–6022.
- Yeh, C.-C., Lin, C.-C., Wang, S.-D., Chen, Y.-S., Su, B.-H., Kao, S.-T., 2006. Protective and anti-inflammatory effect of a traditional Chinese medicine, Xia-Bai-San, by modulating lung local cytokine in a murine model of acute lung injury. *International Immunopharmacology* 6, 1509–1514.